Digital Logic Design ECE 3441 Section 15028 – Spring 2025

Lab #1: TTL Characteristics and SimUAid

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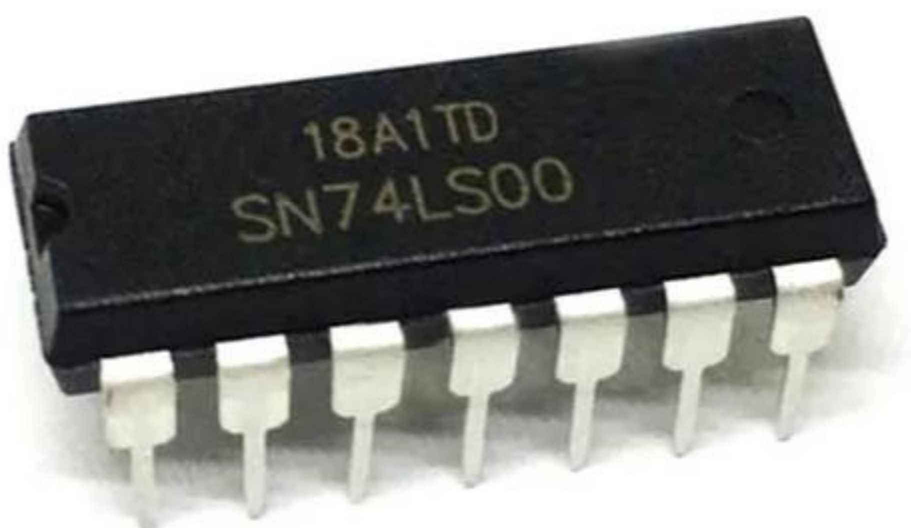
Overview: Goal of the Lab

In this lab, we will learn how to research and read datasheets of electrical components. Students will understand how to pull information from the datasheet and interpret what is being conveyed. They will also learn the basics of packaging and the limitations of the electrical component. Additionally, they will learn the basics of SimUAid or other logic design software to enhance their understanding of digital logic design. Students should walk away with the ability to understand where to understand circuit components and a basic ability to use design software.

Part 1: Getting Familiarized with the 74LS00 TTL Chip

1. What is DIP packaging? Include a figure containing an image of a 74LS00 chip in DIP packaging. Try to make sure the part number is visible! Hint: you have one in your lab kit.



[Figure 1]

DIP packaging is short for Dual inline package. It is the style of packaging that electrical engineers use when implementing integrated circuits. It consists of 2 parallel rows of pins that resemble a spider with many or fewer legs. Most of the time there is a indention of a hole on top to tell the user where the first pin is.

2. There are five fields in the part number SN-74-LS-00-N. Explain what each field tells you about the chip. Applying this breakdown to the chip in your lab kit, how might this influence which datasheet you use, if you found more than one?

1st digits/number “SN” represents the manufacturer’s prefix. Each manufacturer has their own prefix and SN just represents that this part came from Texas Instruments.

2nd set of digits represent the logic family. There are different codes for different families. 74 represents Commercial grade, 54 represents Military grade, and 75 represents interface device.

3rd set of characters represents the sub-family. In this case “LS” stands for low-power Schottky. Which means that it is a type of Schottky diode that is designed to work at low power levels.

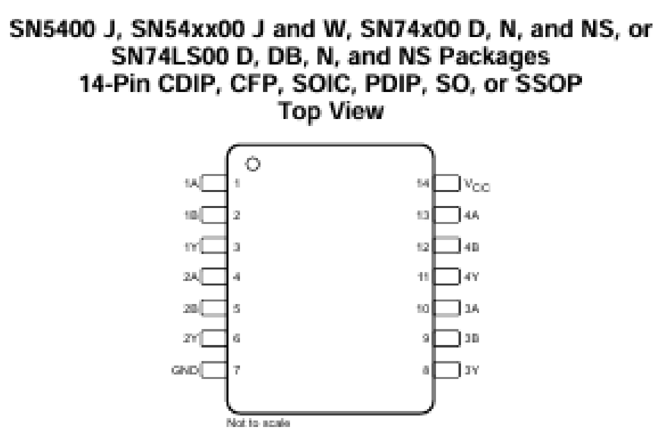
4th set of characters represents the function of the component (IC). In this case “00” represents the function of Integrated Circuit. In this case it is a quad two-input NAND gate.

5th set of characters represents the manufacture’s package type. In this case “N” represents the DIP packaging that can help us understand the IC. If you didn’t want the IC on a DIP then you could sort the part to find one that didn’t have the “N”.

If you find several datasheets then depending on what you want, you would look at the one that you think works for you. For example, if you are working on commercial electronics then you wouldn’t look for a component that has “54”. If you were looking anything but a NAND gate, then you wouldn’t use any component that has “00”. Depending on what your need is, you would use a different component which means the part number would change. From there you would choose the datasheet with the component that fits your purpose.

3. Using the datasheet, how many total inputs and outputs are there in this TTL chip? How many inputs are used to generate each output? Include a figure of the pinout diagram.

There are a total of four independent 2-input NAND gates which means that there are 8 input spaces. Since, there are a total of 12 pins (excluding ground and Vcc) the remaining 4 must be outputs. This gives us 8 input pins and 4 output pins with each output requiring 2 input pins.



[Figure 2]

4. Using the datasheet, what are pins 7 and 14 used for? What does the little notch in the plastic case do?

Pins 7 and 14 are the ground and power pins (Vcc) respectively. The little plastic notch tells us what side to start from so we can get an orientation of what pin 1 and so forth are. If there was no notch, then it is easy to get confused where the pins start and the user could destroy the component.

5. Provide a link to the datasheet you are using for this lab.

https://www.ti.com/lit/ds/symlink/sn7400.pdf?HQS=TI-null-null-alldatasheets-df-pf-SEP-wwe&ts=1738599519082&ref_url=https%253A%252F%252Fwww.alldatasheet.com%252F

Part 2: TTL Characteristics of the 74LS00 TTL Chip

Name:	Value or Range:	Units:
V_{OH_min}	MIN: 2.5	[V]
V_{IH_min}	MIN: 2	[V]
V_{IL_max}	MAX: 0.8	[V]
V_{OL_max}	MAX: 0.5	[V]
V_{CC}	Min: 4.75 NOM: 5 MAX: 5.25	[V]
I_{OL_max}	MAX: 8	[mA]
I_{OH_max}	MAX: -0.4	[mA]
I_{IL_max}	MAX: -0.4	[mA]
I_{IH_max}	MAX:20	[μA]

Questions: (Formulas and calculations must be included)

6. Briefly summarize what each parameter above means.

V_{OH_min} stands for Minimum Output High Voltage, which is the minimum voltage required for the component to register a "1" at that output gate.

V_{IH_min} stands for the minimum Input High voltage which is the minimum voltage level for the component to register a "1" for input.

V_{IL_max} stands for the Maximum Input Low Voltage. This means that this is the maximum voltage allowed for the Input to register a "0" when it goes into the component.

V_{OL_max} stands for the Maximum Output Low Voltage. It means that this is the maximum voltage outputted for a "0" from this component.

V_{CC} means this is the Voltage from the power source. It has a minimum and maximum to give some tolerance. Every value in between this range will cause the component to work.

I_{OL_max} is the maximum current leaving the pins when the pin registers a low logic state.

I_{OH_max} is the maximum current leaving the pin when the pin is outputting a high logic state.
I_{IL_max} is the maximum allowed current entering the input pin for it to register a low state.
I_{IH_max} is the maximum allowed current entering the input pins for it to register a high logic state.

7. What does positive logic and negative logic refer to?

Positive logic refers to the logical system where "1" is counted as a high voltage and "0" is counted as a lower voltage (non-zero most of the time).
Negative logic is the opposite of positive logic. This means that "1" is highlighted by a low voltage and "0" is highlighted by a high voltage.
Depending on the component it might be more efficient to use positive logic and vice versa applies.

8. Explain what fan-in and fan-out mean. Determine the fan-in and calculate the fan-out of a single logic gate in the 74LS00 TTL chip. (Remember: fan-out needs to be calculated for both logical high and logical low outputs, just in case one is smaller). What should you do in your circuit to comply with fan-out?

Fan-in is the maximum number of inputs that the gate can receive. Fan-out is the number of inputs that our original output can reach.
Calculations:
I_{OL_max} = 8mA, I_{IL_max} = -0.4mA, I_{OH_max} = -0.4mA, I_{IH_max} = 20µA
$\boldsymbol{Fan-out_L} = \frac{\mathbf{IOL_{max}}}{\mathbf{IIL_{max}}} = \frac{\mathbf{8}\ [mA]}{\mathbf{-0.4}\ [mA]}$ **= |-20|=20**
$\boldsymbol{Fan-out_H} = \frac{\boldsymbol{IOH}_{\mathbf{max}}}{\mathbf{IIH_{max}}} = \frac{\mathbf{-0.4}\ [mA]}{\mathbf{20}\ [\mu A]}$ **= |-20|=20**
Fan-in = $\#\boldsymbol{of\ inputs\ per\ output}$ **= 2**

You should build the circuit in such a way that you don't overload the component. This can include choosing different components or putting in buffers to ensure that the circuit is safe and reliable. If you are within the Fan limit, then that component won't break due to overload which can help with the longevity of the device you are trying to build.

9. Explain what pull-up and pull-down resistors are. What happens if you don't put them when you use DIP switches as inputs to your TTL chips? (Hint: research "floating input voltage").

Pull-up resistors are components that connect the inputs to the voltage supply to keep the inputs at the "high" voltage level. So if the pin is not connected to something then it automatically sets it to the power supply voltage.
Pull-down resistors work in the opposite. They connect the inputs to the ground keeping the inputs at the "low" level.

If you don't have these components, you can have something called "floating inputs". This means that the input can pick up noise preventing you from getting an accurate reading if it is to be a high or low voltage. It can cause bad logic since now the voltage is fluctuating between the "1" and "0" ranges and can hurt the information being passed.

10. Calculate the maximum pull-up and pull-down resistances for a single logic gate input in the 74LS00 TTL chip using the specifications you put in the table above. Remember: if the resistances are too high, the pull-up/pull-down resistors don't connect the input to ground or 5 [V] well, and you get invalid input voltage levels. Your objective is to find the largest resistance such that the input low voltage stays below VIL_max (for pull downs), and such that the input high voltage stays above VIH_min (for pull-ups). The currents that pass through the inputs are on the table. Watch your units! Show your work and ask for clarification if you need it. After you've finished, explain in your own words why these limits are important.

We can use Ohm's law to find the pull-up resistor. Since the resistor is connected to the power supply and the component can take only 2[V] as an input then we can find the maximum resistance for the pull-up resistor.

$$R_{Pull\_up} = \frac{Vcc - V_{IH}}{I_{IH\_max}} = \frac{5[V] - 2[V]}{20[\mu A]} = 150[k\Omega]$$

The process is the same for the pull-down resistor except instead of the power voltage, we are using ground since that's where the pull-down resistor pulls too.

$$R_{Pull\_down} = \frac{0 - V_{IL}}{I_{IL\_max}} = \frac{0[V] - 0.8[V]}{-0.4[mA]} = 2\,[k\Omega]$$

If you don't have these resistors to pull-up and down, then you are going to create the floating inputs which will cause distortion in your circuit. If you have distortion then your information going into the rest of your circuit will be flawed and you will have an unpredictable component that may not work in the ways you predict it to work.

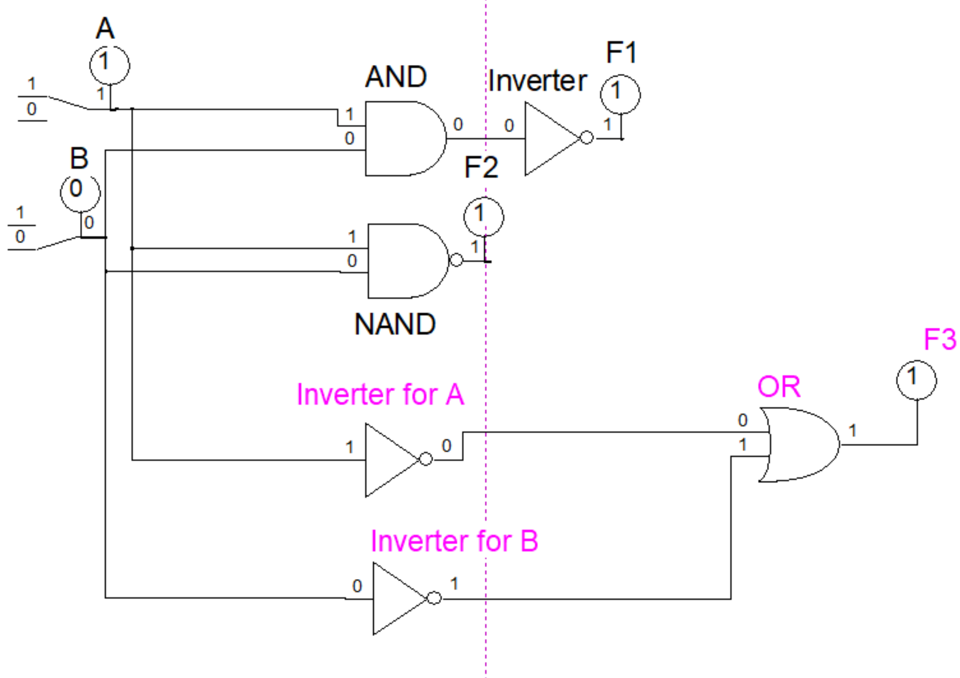
11. In practice, you will connect many logic gate inputs to one pull-up/pull-down resistor. Assuming the characteristics of the 74LS00 TTL chip, consider what happens if you connect n logic gate inputs to a single pull-up/pull-down resistor. Since having n gates multiplies the total current through the resistor by n times, briefly repeat the calculations from Question 10, but with n times the current. How do you modify your pull-up/pull down resistor limit formulas to support n gates?

If we multiply the current by n times, then our modified equation would be like the one of our single components but modifying the current so each component/input can get the correct amount of current. Which would look like the following. In theory we are adding more components that require the same amount of current. However, this results in the resistor becoming smaller to accommodate more components. The same thing would occur for the pull-down resistor.

$$R_{Pull\_up} = \frac{Vcc - V_{IH}}{n * I_{IH\_max}}$$

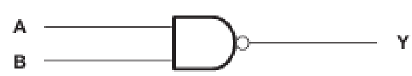
$$R_{Pull_{down}} = \frac{0 - V_{IL}}{n * I_{IL\_max}}$$

Part 3: Software Implementation Using SimUAid



[Figure 3: NAND Gate Simulation in SimUAid]

8.2 Functional Block Diagram



[Figure 4: NAND Gate from Datasheet]

8.4 Device Functional Modes

Table 1 lists the functions of the devices.

Table 1. Functional Table (Each Gate)

INPUTS		OUTPUT
A	B	Y
H	H	L
L	X	H
X	L	H

[Table 1: Truth Table from Datasheet and from Figure 4]

Inputs		Outputs		
A	**B**	**F1**	**F2**	**F3**
0	1	1	1	1
0	0	1	1	1
1	0	1	1	1
1	1	0	0	0

[Table 2: Truth Table for the NAND Gate (75LS00) from SimUAid (Figure 3)]

As we can see from Figure 3 and Table 2, the NAND Gate is the AND Gate + Inverter. We can assume that there is a positive logic flow and "1" is high wile "0" is low. A and B are different inputs that are independent of each other. According to Table 2, F1 = F2=F3, which shows that NAND is just the AND + inverter and the last equivalent circuit.

As seen from Table 1 and Table 2, the Inputs and Outputs line up which means the simulation is the same as what is being conveyed on the datasheet.

Conclusion

The student has learned how to find, understand, and interpret the datasheet of the component. This includes finding maximum current and voltages allowed to safely operate the chip, and what the output voltage and current is expected to be. The student has also learned basic terminology about electrical components like manufacturers, pin locations, and packaging choices. Additionally, they learned how to safely use certain components as well as understanding their component's limitations. They have a basic understanding of how to use other components alongside this one. The student has also learned the basic of SimUAid and how to properly place, edit and simulate certain digital circuits and put that information onto a Truth table.